National Institute of Technology
Rourkela -769008

Department	Computer Science and Engineering
Examination	Mid/End Semester
Program	B.Tech. 6th Semester
Session	Autumn/Spring
Course Code & Course Name	Computer Vision (CS3303)
Marks	30
Duration	120 Minutes

Instructions: ***1. In case of any doubt, write your assumptions, write it clearly and continue.***
2. There are 05 questions printed on two pages of the paper. Attempt all the questions.

*CO: Course Outcome; **BL: Bloom's Taxonomy Level.

Q. No.	Questions	*Marks*	*CO**	*BL***
1. (A)	What is computer Vision? Explain three major milestones in Computer Vision History.	*[2]*	***CO1***	Remember Understand
(B)	What do you mean by Image Magnification? Mathematically, magnification is represented from a projection point of view. In a single-lens system, if the object distance is 20 m and the image distance is 10 mm, find the magnification.	*[4]*		
2. (A)	Derive Affine transform formulae (in matrix form) for Scaling, Translation, and Rotation operations. Get the transformation matrix for the following 1. An image to be zoomed up by a factor of 2 in the x-direction and 3 in the y-direction. 2. An object in an image has to be moved from the original centroid position of P (125, 200) to another position P´ (145, 250). 3. An object in a given image is to be rotated by an angle of $\pi/4$	*[3]* *(1x3)*	***CO2***	Apply Remember
(B)	What are the basic differences between the RGB colour model and the HSI colour model in terms of human perceptions? Describe the conversion formula of these colour models.	*[3]*		
3. (A)	In a given application, an averaging mask is applied to input images to reduce noise, and then a Laplacian mask is applied to enhance small details. Would the result be the same if the order of these operations were reversed? Explain.	*[2]*	***CO2***	Understand Evaluate
(B)	Consider a grayscale image with the following normalised histogram (probability distribution of intensity levels):	*[4]*		

Intensity Level	**0**	**1**	**2**	**3**
Probability	0.1	0.3	0.4	0.2

You need to match this histogram to a specified target histogram:

Intensity Level	**0**	**1**	**2**	**3**
Probability	0.2	0.2	0.3	0.3

Perform histogram specification and determine the mapping of input intensity levels to target intensity levels.

Q4. Consider the convolutional neural network defined by the layers in the left column below. Fill in the shape of the output volume and the number of parameters at each layer. You can write the activation shapes in the format (H, W, C), where H, W, C are the height, width and channel dimensions, respectively. Unless specified, assume padding 1, stride 1 where appropriate. ***[6]*** ***CO3*** Analyse

Notation:

- CONVx-N denotes a convolutional layer with N filters with height and width equal to x.
- POOL-n denotes a n×n max-pooling layer with a stride of n and 0 padding.
- FLATTEN flattens its inputs, identical to torch.nn.flatten / tf.layers.flatten
- FC-N denotes a fully connected layer with N neurons

Layer	Activation Volume Dimensions	Number of Parameters
Input	64 X 64 X 3	0
CONV3-16		
ReLU		
POOL-2		
FLATTEN		
FC-5		
FC-3		

5. (A) Explain the weight-updating equations for the following algorithms: ***[3]***

1. Gradient Descent with momentum
2. Nesterov Accelerated Gradient Descent
3. AdaGrad

(B) Consider a fully connected network with n inputs $x_1, x_2, \ldots x_n$. Suppose there is a single hidden layer with k neurons having f as the activation function and b_j as the bias of *jth* hidden neuron. Further, the output layer contains a single neuron with g as the activation function and B as the bias. Assume that the weights between inputs to the hidden layer are denoted by w and weights between hidden to output layer are denoted as $\boldsymbol{v}$. Write down the output y of the network as a function of $\mathbf{x}$, weights and biases. Suggest the loss function for the regression and classification problems. ***[3]*** ***CO3*** Remember Apply

End of Paper